30 Days Generative AI Roadmap (Structured Schedule)

PHASE 1 – Foundation (Days 1–10)

Day	Focus Area	Topics to Cover	Practice Task
1	Introduction to AI	What is AI? AI applications	Write real-life AI examples (education, business)
2	ML vs DL	Machine Learning vs Deep Learning	Compare ML vs DL in table format
3	Learning Types	Supervised vs Unsupervised learning	Identify real-world examples
4	ML Problems	Classification & Regression	Try simple sklearn model
5	Model Issues	Overfitting & Underfitting	Plot simple training vs testing graph
6	Evaluation	Accuracy, Precision, Recall, F1-score	Calculate metrics using sklearn
7	Python for AI	NumPy basics	Matrix operations
8	Data Handling	Pandas	Load CSV & analyze
9	Visualization	Matplotlib	Plot graph from dataset
10	ML Practice	Scikit-learn	Build simple classification model

PHASE 2 – Deep Learning (Days 11–14)

Day	Focus Area	Topics	Practice
11	Neural Networks	Perceptron	Implement simple perceptron
12	Training Process	Backpropagation	Understand gradient descent visually
13	Activation Functions	ReLU, Sigmoid, Softmax	Plot activation functions
14	Framework	PyTorch (recommended)	Build simple neural network

PHASE 3 – NLP (Days 15–19)

Day	Focus Area	Topics	Practice
15	NLP Basics	Text preprocessing	Clean sample text
16	Tokenization	Word & sentence tokenization	Tokenize paragraph
17	Word Embeddings	Word2Vec, GloVe	Generate embeddings
18	Sequential Models	RNN & LSTM	Understand architecture
19	Attention	Attention mechanism	Why transformers replaced LSTM? (Write explanation)

PHASE 4 – Transformers & LLMs (Days 20–23)

Day	Focus Area	Topics	Practice
20	Transformer	Transformer architecture	Draw diagram
21	Self-Attention	Encoder vs Decoder	Explain with example
22	GPT	GPT architecture	Generate text using API
23	BERT	BERT architecture	Compare GPT vs BERT

PHASE 5 – Prompt Engineering (Days 24–25)

Day	Focus Area	Topics	Practice
24	Prompt Basics	Zero-shot, Few-shot, Role prompting	Create 5 different prompt styles
25	Advanced Prompting	Chain-of-thought, Temperature control	Generate structured JSON output

PHASE 6 – GenAI APIs (Days 26–27)

Day	Focus Area	Topics	Practice Project
26	API Usage	OpenAI, Gemini, HuggingFace	Build AI Content Generator
27	Application Build	Connect API to UI	Build AI Quiz Generator or Resume Builder

PHASE 7 – RAG (Days 28–29)

Day	Focus Area	Topics	Practice
28	Embeddings & Vector DB	FAISS / Pinecone	Store & search text chunks
29	RAG Implementation	PDF → Q&A bot	Build simple document chatbot

PHASE 8 – Fine-Tuning (Day 30)

Day	Focus Area	Topics	Practice
30	Fine-Tuning	LoRA, PEFT, Custom training	Study concept + small demo